

FINAL TERM EXAMINATION

Fall 2009

CS301- Data Structures

Time: 120 min

Marks: 75

Question No: 1 (Marks: 1) - Please choose one

The arguments passed to a function should match in number, type and order with the parameters in the function definition.

[▶ True](#)

▶ False

Question No: 2 (Marks: 1) - Please choose one

If numbers 5, 222, 4, 48 are inserted in a queue, which one will be removed first?

▶ 48

▶ 4

▶ 222

[▶ 5](#)

Question No: 3 (Marks: 1) - Please choose one

Suppose currentNode refers to a node in a linked list (using the Node class with member variables called data and nextNode). What statement changes currentNode so that it refers to the next node?

▶ currentNode ++;

▶ currentNode = nextNode;

▶ currentNode += nextNode;

▶ currentNode = currentNode->nextNode;

Question No: 4 (Marks: 1) - Please choose one

A **Compound Data Structure** is the data structure which can have multiple data items of same type or of different types. Which of the following can be considered compound data structure?

▶ Arrays

[▶ LinkLists](#)

- ▶ Binary Search Trees
- ▶ All of the given options

Question No: 5 (Marks: 1) - Please choose one

Here is a small function definition:

```
void f(int i, int &k)
{
    i = 1;
    k = 2;
}
```

Suppose that a main program has two integer variables x and y , which are given the value 0. Then the main program calls $f(x, y)$; What are the values of x and y after the function f finishes?

- ▶ Both x and y are still 0.
- ▶ x is now 1, but y is still 0.
- ▶ [x is still 0, but y is now 2.](#)
- ▶ x is now 1, and y is now 2.

Question No: 6 (Marks: 1) - Please choose one

The difference between a binary tree and a binary search tree is that ,

- ▶ a binary search tree has two children per node whereas a binary tree can have none, one, or two children per node
- ▶ in binary search tree nodes are inserted based on the values they contain
- ▶ in binary tree nodes are inserted based on the values they contain
- ▶ [none of these](#)

Question No: 7 (Marks: 1) - Please choose one

Compiler uses which one of the following to evaluate a mathematical equation,

- ▶ Binary Tree
- ▶ Binary Search Tree
- ▶ [Parse Tree](#)
- ▶ AVL Tree

Question No: 8 (Marks: 1) - Please choose one

If there are 56 internal nodes in a binary tree then how many external nodes this binary tree will have?

- ▶ 54
- ▶ [55](#)
- ▶ 56
- ▶ 57

Question No: 9 (Marks: 1) - Please choose one

If there are 23 external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

- ▶ 23
- ▶ 24
- ▶ 21
- ▶ [22](#)

Question No: 10 (Marks: 1) - Please choose one

Which of the following method is helpful in creating the heap at once?

- ▶ insert
- ▶ add
- ▶ update
- ▶ [preculatDown \(lecture #31 page 1\)](#)

Question No: 11 (Marks: 1) - Please choose one

The definition of Transitivity property is

- ▶ For all element x member of S , $x R x$
- ▶ For all elements x and y , $x R y$ if and only if $y R x$
- ▶ [For all elements \$x, y\$ and \$z\$, if \$x R y\$ and \$y R z\$ then \$x R z\$](#)
- ▶ For all elements w, x, y and z , if $x R y$ and $w R z$ then $x R z$

Question No: 12 (Marks: 1) - Please choose one

A binary tree of N nodes has _____.

- ▶ $\log_{10} N$ levels
- ▶ [Log₂ N levels](#)
- ▶ $N / 2$ levels
- ▶ $N \times 2$ levels

Question No: 13 (Marks: 1) - Please choose one

If there are N elements in an array then the number of maximum steps needed to find an element using Binary Search is _____ .

- ▶ N
- ▶ N^2
- ▶ $N \log_2 N$
- ▶ [log₂N](#)

Question No: 14 (Marks: 1) - Please choose one

Consider the following array

23 15 5 12 40 10 7

After the first pass of a particular algorithm, the array looks like

15 5 12 23 10 7 40

Name the algorithm used

- ▶ Heap sort
- ▶ Selection sort
- ▶ Insertion sort
- ▶ [Bubble sort](#)

Question No: 15 (Marks: 1) - Please choose one

If both pointers of the node in a binary tree are NULL then it will be a/an _____ .

- ▶ Inner node
- ▶ [Leaf node](#)
- ▶ Root node
- ▶ None of the given options

Question No: 16 (Marks: 1) - Please choose one

By using _____ we avoid the recursive method of traversing a Tree, which makes use of stacks and consumes a lot of memory and time.

- ▶ Binary tree only
- ▶ [Threaded binary tree](#)
- ▶ Heap data structure

- ▶ Huffman encoding

Question No: 17 (Marks: 1) - Please choose one

A complete binary tree of height 3 has between _____ nodes.

- ▶ 8 to 14
- ▶ [8 to 15](#)
- ▶ 8 to 16
- ▶ 8 to 17

Question No: 18 (Marks: 1) - Please choose one

Consider a min heap, represented by the following array:

3,4,6,7,5,10

After inserting a node with value 1. Which of the following is the updated min heap?

- ▶ 3,4,6,7,5,10,1
- ▶ 3,4,6,7,5,1,10
- ▶ 3,4,1,5,7,10,6
- ▶ [1,4,3,5,7,10,6](#) close to correct but correct ans is 1,4,3,7,5,10,6

Question No: 19 (Marks: 1) - Please choose one

Consider a min heap, represented by the following array:

10,30,20,70,40,50,80,60

After inserting a node with value 31. Which of the following is the updated min heap?

- ▶ [10,30,20,31,40,50,80,60,70](#)
- ▶ 10,30,20,70,40,50,80,60,31
- ▶ 10,31,20,30,40,50,80,60,31
- ▶ 31,10,30,20,70,40,50,80,60

Question No: 20 (Marks: 1) - Please choose one

Which one of the following algorithms is most widely used due to its good average time,

- ▶ Bubble Sort
- ▶ Insertion Sort
- ▶ [Quick Sort](#)
- ▶ Merge Sort

Question No: 21 (Marks: 1) - Please choose one

Which of the following statement is correct about find(x) operation:

- ▶ A find(x) on element x is performed by returning exactly the same node that is found.
- ▶ A find(x) on element x is performed by returning the root of the tree containing x.
- ▶ A find(x) on element x is performed by returning the whole tree itself containing x.
- ▶ A find(x) on element x is performed by returning TRUE.

Question No: 22 (Marks: 1) - Please choose one

Which of the following statement is NOT correct about find operation:

- ▶ It is not a requirement that a find operation returns any specific name, just that finds on two elements return the same answer if and only if they are in the same set.
- ▶ One idea might be to use a tree to represent each set, since each element in a tree has the same root, thus the root can be used to name the set.
- ▶ Initially each set contains one element.
- ▶ Initially each set contains one element and it does not make sense to make a tree of one node only.

Question No: 23 (Marks: 1) - Please choose one

The following are statements related to queues.

- (i) The last item to be added to a queue is the first item to be removed **False statement**
- (ii) A queue is a structure in which both ends are not used **False statement**
- (iii) The last element hasn't to wait until all elements preceding it on the queue are removed **False statement**
- (iv) A queue is said to be a last-in-first-out list or LIFO data structure. **False statement**

Which of the above is/are related to normal queues?

- ▶ (iii) and (ii) only
- ▶ (i), (ii) and (iv) only
- ▶ (ii) and (iv) only
- ▶ [None of the given options](#)

Question No: 24 (Marks: 1) - Please choose one

The maximum number of external nodes (leaves) for a binary tree of height H is

-
- ▶ [2^H](#)
 - ▶ 2^H+1
 - ▶ 2^H-1
 - ▶ 2^H+2

Question No: 25 (Marks: 1) - Please choose one

In complete binary tree the bottom level is filled from _____

- ▶ Left to right
- ▶ Right to left
- ▶ Not filled at all
- ▶ None of the given options

Question No: 26 (Marks: 1) - Please choose one

We are given N items to build a heap , this can be done with _____ successive inserts.

- ▶ N-1
- ▶ [N](#)
- ▶ N+1
- ▶ N²

Question No: 27 (Marks: 1) - Please choose one

Suppose we had a hash table whose hash function is " $n \% 12$ ", if the number 35 is already in the hash table, which of the following numbers would cause a collision?

- ▶ 144
- ▶ 145
- ▶ 143
- ▶ 148

Question No: 28 (Marks: 1) - Please choose one

Here is an array of ten integers:

5 3 8 9 1 7 0 2 6 4

The array after the FIRST iteration of the large loop in a selection sort (sorting from smallest to largest).

- ▶ 0 3 8 9 1 7 5 2 6 4
- ▶ 2 6 4 0 3 8 9 1 7 5
- ▶ 2 6 4 9 1 7 0 3 8 5
- ▶ 0 3 8 2 6 4 9 1 7 5

Question No: 29 (Marks: 1) - Please choose one

What requirement is placed on an array, so that *binary search* may be used to locate an entry? ▶ The array elements must form a heap.

- ▶ The array must have at least 2 entries.

▶ The array must be sorted.

- ▶ The array's size must be a power of two.

Question No: 30 (Marks: 1) - Please choose one

In case of deleting a node from AVL tree, rotation could be prolong to the *root* node.

► Yes

► No

Question No: 31 (Marks: 1)

In merge sort do we need to have extra memory, justify your answer in either case.

Yes we need extra memory in merge sort.

Question No: 32 (Marks: 1)

Where is Inorder Predecessor of a non leaf node is present in a Binary Search Tree?

Question No: 33 (Marks: 2)

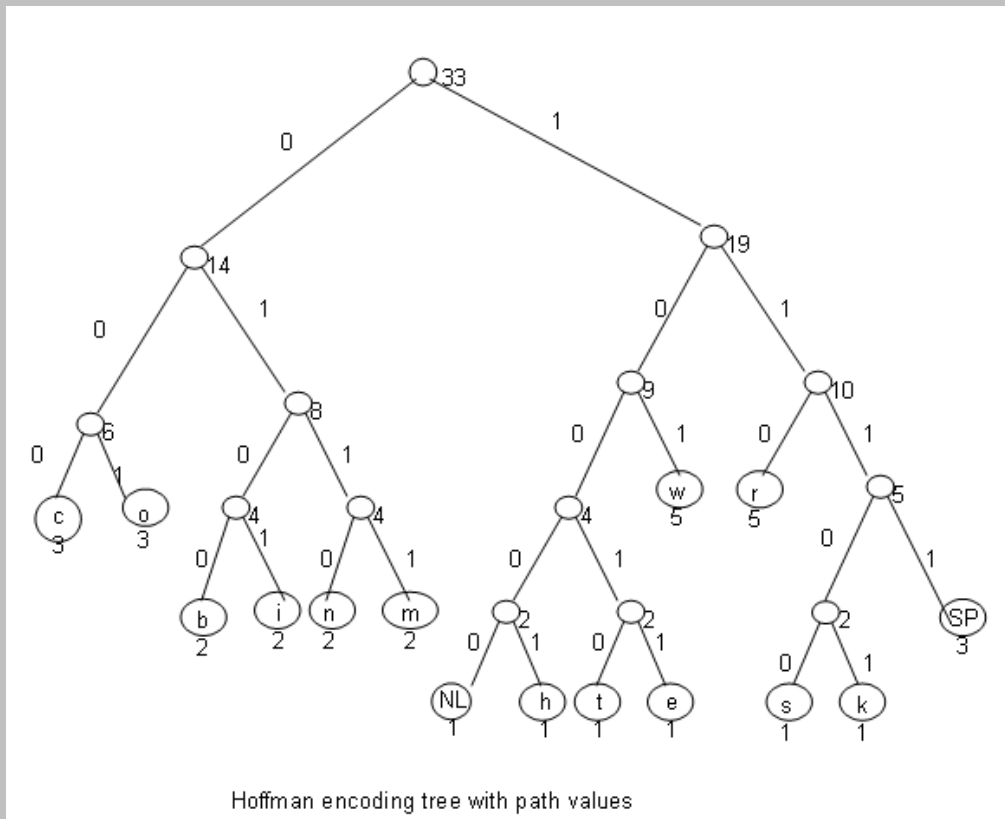
How we can search an element in Skip List.

Question No: 34 (Marks: 2)

What is the drawback of using arrays to store Binary Search Trees.

Question No: 35 (Marks: 3)

Calculate the codes of the following characters in table below using the hoffman encoding tree,



character	Code
NL	10000
SP	1111
o	001
b	0100
i	0101
r	110

Question No: 36 (Marks: 3)

"For smaller lists, linear insertion sort performs well, but for larger lists, quick sort is suitable to apply." Justify why?

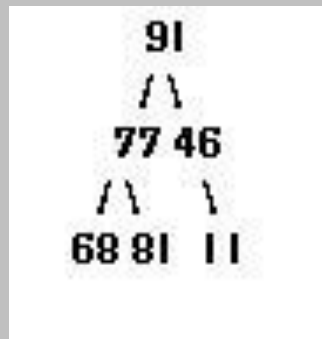
Question No: 37 (Marks: 3)

Suppose that we have implemented a priority queue by storing the items in a heap. We are now executing a reheapification downward and the out-of-place node has priority of 42. The node's parent has a priority of 72, the left child has priority 52 and the node's right child has priority 62. Which statement best describes the status of the reheapification.

- A. The reheapification is done.
- B. The next step will swap the out-of-place node with its parent.
- C. The next step will swap the out-of-place node with its left child.
- D. The next step will swap the out-of-place node with its right child.
- E. None of these.

Question No: 38 (Marks: 5)

Give two different reasons to explain why the following binary tree is not a heap:



Question No: 39 (Marks: 5)

Here is an array of ten integers:

5 3 8 9 1 7 0 2 6 4

Sort the array by using selection sort algorithm and show content of array after each step.

	5	3	8	9	1	7	0	2	6	4
--	---	---	---	---	---	---	---	---	---	---

STEP 1	0	3	8	9	1	7	5	2	6	4
STEP 2	0	1	8	9	3	7	5	2	6	4
STEP3	0	1	2	9	3	7	5	8	6	4
STEP 4	0	1	2	3	9	7	5	8	6	4
STEP5	0	1	2	3	4	7	5	8	6	9
STEP6	0	1	2	3	4	5	7	8	6	9
STEP 7	0	1	2	3	4	5	6	8	7	9
STEP8	0	1	2	3	4	5	6	7	8	9
	0	1	2	3	4	5	6	7	8	9

Question No: 40 (Marks: 10)

A long sequence of vowels needs to be transmitted efficiently so a programmer decides to use Huffman encoding to encode the vowels. A distribution count study of typical data yielded the following frequency table.

Frequency Table		
character	frequency	Huffman code
A	33978	-----
E	20676	-----
I	15814	-----
O	21552	-----
U	10324	-----
Y	4975	-----

- Create a Huffman tree to determine the binary codes for each character.
- Fill the codes into the table above.
- Encode the following sequence EIYOUA

Question No: 41 (Marks: 10)

Consider the following tree.

- Show that either it is a heap or not.
- If it is a heap then what type of heap is it?
- Add 40 in the heap and convert it in max heap.

